BCS-073	NEURAL NETWORK	L T P 3 0 0	3 Credits

Course Objectives:

CO1	To understand the basic concepts of learning and decision trees.
CO2	To understand the neural networks and genetic algorithms
CO3	To understand the instant based learning
CO4	To understand the analytical learning and reinforced learning
CO5	To learn about SOM

Detailed Syllabus

MODULE-I

Neuro computing and Neuroscience: Historical notes, human Brain, neuron Mode l, Knowledge representation, Al and NN. Learning process: Supervised and unsupervised learning, Error correctionlearning, competitive learning, adaptation, statistical nature of the learning process.

Data processing: Scaling, normalization, Transformation (FT/FFT), principal component analysis, regression, covariance matrix, eigen values & eigen vectors. Basic Models of Artificial neurons, activation Functions, aggregation function, single neuron computation, multilayer perceptron, least mean square algorithm, gradient descent rule, nonlinearly separable problems and bench mark problems in NN.

MODULE-II

Multilayered network architecture, back propagation algorithm, heuristics for making BP algorithm performs better. Accelerated learning BP (like recursive least square, quick prop, PROP algorithm), approximation properties of RBF networks and comparison with multilayer perceptran.

MODULE-III

Recurrent network and temporal feed-forward network, implementation with BP, self organizing map and SOM algorithm, properties of feature map and computer simulation. Principal component and Independent component analysis, application to image and signal processing.

Complex valued NN and complex valued BP, analyticity of activation function, application in2D information processing. Complexity analysis of network models. Soft computing. Neuro-Fuzzy-genetic algorithm Integration.

CO1	Choose the learning techniques with this basic knowledge.
CO2	Apply effectively neural networks and genetic algorithms for appropriate applications.
CO3	Explain the different machine learning techniques
CO4	Choose and differentiate reinforcement and analytical learning techniques
CO5	Can implement about SOM

Text Books:

1. J.A. Anderson, AnIntoduction to Neural Networks, MIT
2. Hagen Demuth Beale, Neural Network Design, Cengage Learning

References:

1. R.L. Harvey, Neural Network Principles, PHI
2. Kosko, Neural Network and Fuzzy Sets, PHI